MEAN SHIFT IN COMPUTER VISION

Detailed Explanation Using Pixels – Simple Words

1. What is Mean Shift?

Mean Shift is an algorithm used in computer vision to find groups or regions where similar pixels are concentrated. It can be used for image segmentation, object tracking, finding regions with similar colours, and finding the centre of a group of data points.

Easy meaning: Mean = average, Shift = move. Therefore, Mean Shift means **calculate the average position and move toward it**.

2. Very Small Image Example

B	**B**	**B**	**B**	**B**
B	**R**	**R**	**R**	**B**
B	**R**	**R**	**R**	**B**
B	**R**	**R**	**R**	**B**
B	**B**	**B**	**B**	**B**

B = blue background
R = red object

3. A Pixel Has Position and Colour

A pixel is not only a colour. It also has a location in the image. For example, a pixel may be at position **(x = 3, y = 3)** and have a red colour represented approximately as **(R=200, G=30, B=30)**.

Conceptual pixel information: (x, y, R, G, B)

4. How Mean Shift Starts

Suppose we select one pixel, for example **P = (3,3)**. Mean Shift creates a local search window around this pixel. The window contains nearby pixels that may belong to the same region.

Important: The actual image pixels are not physically moved. Mean Shift moves the **search location/window** toward a dense region.

5. Step 1 – Choose a Pixel

Choose a starting pixel. Example: **P = (3,3)**.

6. Step 2 – Look at Nearby Pixels

Mean Shift checks pixels close to the selected pixel. It does not blindly use every pixel in the whole image. It focuses on a local neighbourhood.

7. Step 3 – Compare Colour Similarity

Mean Shift considers whether nearby pixels have similar colours. Pixels that are close together and have similar colour values are more likely to belong to the same region.

Two important ideas:
• Spatial closeness – How close are the pixels in the image?
• Colour similarity – How similar are their colours?

8. Step 4 – Calculate the Mean Position

Suppose the selected red pixels are:
(2,2), (3,2), (4,2), (2,3), (3,3), (4,3), (2,4), (3,4), (4,4).

Mean X: (2+3+4+2+3+4+2+3+4) / 9 = 3
Mean Y: (2+2+2+3+3+3+4+4+4) / 9 = 3

Mean position = (3,3)

If the current search location is already (3,3), there is no movement. The point has reached a stable region.

9. Edge Pixel Example – Actual Shift

Now choose an edge pixel **(2,2)**. Suppose the nearby similar red pixels are:

(2,2), (3,2), (2,3), (3,3)

Mean X: (2+3+2+3)/4 = 2.5
Mean Y: (2+2+3+3)/4 = 2.5

Mean = (2.5, 2.5)

Current location = (2,2). Therefore the search location shifts toward (2.5,2.5).

10. Repeat the Process

```
Start:        (2,2)
First shift:  (2,2) → (2.5,2.5)
Next shift:   (2.5,2.5) → (2.8,2.8)
Next shift:   (2.8,2.8) → (3.0,3.0)
Stable:       (3.0,3.0)
```

When the movement becomes very small, Mean Shift stops. This is called **convergence**.

11. Mean Shift Formula – Simple Meaning

Mean Shift = Mean position – **Current position**

For example, if current position = (2,2) and mean position = (2.5,2.5), then:

```
Shift = (2.5, 2.5) – (2, 2)
      = (0.5, 0.5)
```

12. What is a High-Density Region?

Imagine many similar pixels are concentrated in one area. That area has a high density of similar data points. Mean Shift naturally moves the search location toward this concentration.

Simple idea: Many similar pixels together → high density → Mean Shift moves toward them.

13. Feature Space – Simple Explanation

Feature space means the information we use to describe a pixel. For a simple colour image, we can think of a pixel as:

```
(x, y, R, G, B)
```

Here x and y describe location, while R, G and B describe colour. Mean Shift can therefore consider both where a pixel is and how it looks.

14. Complete Flow of Mean Shift

1	Choose a pixel
↓	Create a local search window
↓	Find nearby pixels
↓	Compare their colours
↓	Select similar nearby pixels
↓	Calculate their average position
↓	Move the search location toward the average
↓	Look again
↓	Repeat the process
↓	Movement becomes very small
STOP	Convergence / stable region found

15. Simple OpenCV Code

```
import cv2

img = cv2.imread("image.jpg")

result = cv2.pyrMeanShiftFiltering(
    img,
    21,
    51
)

cv2.imshow("Original", img)
cv2.imshow("Mean Shift", result)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

16. Line-by-Line Code Explanation

Line	Simple meaning
import cv2	Loads OpenCV.
img = cv2.imread("image.jpg")	Reads the input image.
cv2.pyrMeanShiftFiltering(...)	Applies Mean Shift filtering to the image.
21	Spatial parameter (sp): controls the neighbourhood in image space.
51	Colour parameter (sr): controls the allowed colour range.
cv2.imshow("Original", img)	Displays the original image.
cv2.imshow("Mean Shift", result)	Displays the processed image.
cv2.waitKey(0)	Waits for a key press.
cv2.destroyAllWindows()	Closes the OpenCV windows.

17. Easy Memory Trick for sp and sr

sp → **spatial** → where the pixel is
sr → **range / colour similarity** → how similar the colour is

18. Mean Shift Does NOT Move the Actual Image Pixels

This is an important point. Mean Shift does not take a pixel and physically move it to another location. Instead, it moves the **search window or search point in feature space** toward the region where similar pixels are concentrated.

19. Simple Real-Life Analogy

Imagine people standing in a large room. You stand at one position and look at people near you who are similar to the group you are interested in. You calculate the average location of those people and walk toward that average. You look again, calculate another average, and move again. Eventually you reach the centre of the group. That is the basic idea of Mean Shift.

20. Mean Shift vs. K-Means – Quick Difference

Feature	Mean Shift	K-Means
Clusters	Finds dense regions automatically	Usually requires number of clusters K
Main idea	Move toward mean of nearby points	Assign points to nearest centroid
Use	Tracking, segmentation, density-based grouping	General clustering/colour grouping

21. Complete Pixel-Based Flow in One View

```
PIXEL
  ↓
Find nearby pixels
  ↓
Check colour similarity
```

```
  ↓
Keep similar nearby pixels
  ↓
Calculate their average position
  ↓
Compare average with current position
  ↓
SHIFT toward the average
  ↓
Look again
  ↓
Calculate average again
  ↓
SHIFT again
  ↓
Repeat
  ↓
Movement becomes very small
  ↓
STOP → stable region
```

22. Easy Memory Line

LOOK → COMPARE → AVERAGE → SHIFT → REPEAT

23. One-Line Definition for Exam

Mean Shift is an iterative computer vision technique that moves a search window toward the average location of similar nearby pixels or data points until it reaches a stable high-density region.

24. Final Concept in Very Simple Words

Mean Shift starts from a pixel, looks around it, finds nearby pixels that are similar, calculates where those similar pixels are concentrated, moves toward that concentration, and repeats the process until the position becomes stable.